



Retriever Performance Report

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Dit rapport vergelijkt 5 retrievers voor retrieval-augmented generation. Het geeft een samenvatting van de corpus- en queryset, vergelijkt de rankingkwaliteit en runtime, en toont vervolgens de configuratie en voorbeeldresultaten van elke retriever. Gebruik de vergelijkingsectie om te bepalen welke retriever het meest waarschijnlijk nuttig bewijs in het contextvenster van de generator plaatst.

De gesynthetiseerde retrieval-dataset is ontworpen om de prestaties van retrieval-methoden binnen Retrieval-Augmented Generation (RAG) te evalueren. Het corpus bevat technische documenten en tekstchunks uit Symfony-documentatie, waarin uitleg wordt gegeven over configuratie, implementatie en probleemoplossing binnen een PHP-ontwikkelomgeving. Deze documenten bevatten informatie over onderwerpen zoals beveiliging, caching, event dispatching en gebruikersbeheer. De gegenereerde vragen zijn realistische zoekopdrachten die ontwikkelaars zouden stellen wanneer ze een bepaald probleem oplossen, zoals het implementeren van een beveiligde admin-gebied of het cache-erend van API-aanroepen. De vragen zijn opgesteld met een focus op doelen en problemen in plaats van specifieke componenten, waardoor ze een goede test vormen voor de relevantie van documenten. De dataset wordt gebruikt om te meten hoe goed retrieval-systemen relevante informatie kunnen selecteren en is bedoeld om de kwaliteit van zoekresultaten te beoordelen op basis van de mate waarin documenten direct helpen bij het oplossen van de ontwikkelaarsproblemen die in de vragen zijn beschreven.

Machine Summary

Machine Summary

System	Linux
Processor	x86_64
Cores	32
Ram (Gb)	377.56
Python Version	3.12.3
Cuda	Available (2 GPU(s))
Gpu Name	NVIDIA RTX A5000

Corpus Summary

Corpus and Queries

Documents	2,594
Documents With Title	0
Avg Document Chars	695.1766
Min Document Chars	5
Max Document Chars	1,000
Queries	101
Avg Query Chars	67.6733
Min Query Chars	45

Max Query Chars	102
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Sample Query

- how can I limit login attempts to prevent brute force attacks
- how can I check for missing or unused translations in my Symfony project
- how can I split code into separate bundles for better performance
- how can I manipulate strings by joining, splitting, truncating, and reversing them
- how can I configure multiple bundles in a single file

Retriever Comparison

Retrieval and Rerank Metrics

Retriever	NDCG@10	Recall@10
BM25Retriever (bm25)	0.5017	0.5129
BM25Retriever (bm25) + rerank	0.6622	0.6287
SpladeRetriever (splade) + rerank	0.6022	0.5584
DenseRetrieverSentenceBert (dense)	0.7091	0.7446
DenseRetrieverSentenceBert (dense) + rerank	0.77	0.7703
HybridRetriever (hybrid)	0.7087	0.7644
HybridRetriever (hybrid) + rerank	0.77	0.7703
LinearRagRetriever (linear)	0.6952	0.7292
LinearRagRetriever (linear) + rerank	0.7581	0.7495

Chart Colors

Series
Base retrieval
Rerank



Uitleg metrics: Deze scores evalueren de retrievalfase voordat het taalmodel een antwoord genereert. Binnen een RAG-pipeline betekent betere retrieval dat de generator relevanter en beter gerankt bewijs ontvangt en minder hoeft te vertrouwen op niet-onderbouwde modelkennis.

Recall@k meet dekking: hoeveel relevante documenten zijn gevonden binnen de top k. **Precision@k** meet focus: hoeveel van de top k daadwerkelijk relevant is in plaats van afleidende context. **MAP@k** belooft relevante documenten die consistent vroeg verschijnen over meerdere queries heen. **NDCG@k** belooft zowel de rangorde als graduele relevantie en is daarom vaak de beste enkele indicator voor de vraag of het sterkste bewijs als eerste de prompt bereikt.

Praktische interpretatie: Gebruik deze resultaten niet alleen om de hoogste score te kiezen, maar om te bepalen welke retriever het beste past bij het doel van de RAG-toepassing. Een hoge Recall@k is vooral belangrijk wanneer het systeem zo min mogelijk relevante informatie mag missen, bijvoorbeeld bij juridische of ondersteunende zoekvragen. Een hoge Precision@k is belangrijk wanneer het contextvenster beperkt is en irrelevante passages de generator kunnen afleiden. Wanneer reranking duidelijk betere NDCG- of MAP-scores geeft, betekent dit dat de juiste documenten niet alleen gevonden worden, maar ook hoger in de ranglijst komen te staan. Dat is praktisch waardevol omdat een taalmodel meestal sterker leunt op de eerste passages in de context. Runtime moet daarnaast worden meegewogen: een iets lagere score kan acceptabel zijn wanneer de retriever veel sneller is of eenvoudiger te onderhouden blijft.

Practical Limitations and Runtime Trade-offs

Retriever	Query Time	Queries/Sec	Index Time	Document/Sec	Storage
BM25Retriever (bm25)	0.5180s	194.97	4.4370s	584.626	2.00MB
BM25Retriever (bm25) + rerank	8.9959s	11.2323	4.4370s	584.626	2.00MB
SpladeRetriever (splade)	1.1345s	89.0287	56.4633s	45.9414	26.44MB
SpladeRetriever (splade) + rerank	9.4931s	10.646	56.4633s	45.9414	26.44MB
DenseRetrieverSentenceBert (dense)	0.0090s	11232.4526	0.0039s	662083.1235	10.62MB
DenseRetrieverSentenceBert (dense) + rerank	8.2746s	12.2116	0.0039s	662083.1235	10.62MB
HybridRetriever (hybrid)	0.5287s	191.0499	4.4409s	584.1102	12.63MB
HybridRetriever (hybrid) + rerank	8.8049s	11.4765	4.4409s	584.1102	12.63MB
LinearRagRetriever (linear)	12.7062s	7.9489	13.2822s	195.2992	2.29GB
LinearRagRetriever (linear) + rerank	20.9271s	4.8262	13.2822s	195.2992	2.29GB

Quality versus Query Throughput

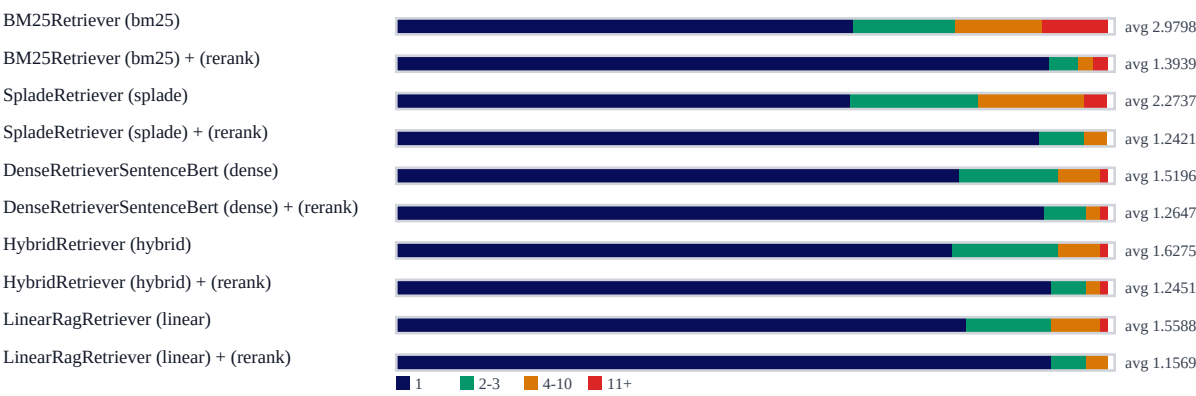


Uitleg runtime en throughput: Deze resultaten evalueren de praktische prestaties van de retrievalfase naast retrievalkwaliteit. Query time meet hoeveel tijd nodig is om queries te verwerken, terwijl index time aangeeft hoeveel tijd nodig is om de volledige index op te bouwen. Queries/sec en documents/sec meten de throughput van het systeem en geven inzicht in schaalbaarheid binnen grotere RAG-omgevingen. Storage MB toont hoeveel opslagruimte de retrievalindex gebruikt, wat belangrijk kan zijn binnen productieomgevingen waar infrastructuurkosten en geheugenverbruik een rol spelen. De grafiek combineert retrievalkwaliteit met throughput door Recall@10 af te zetten tegen queries per seconde. Hogere posities betekenen betere retrievaldekking, terwijl posities verder naar rechts wijzen op hogere snelheid en lagere latency.

Praktische interpretatie: Gebruik deze resultaten niet alleen om de snelste of meest accurate retriever te kiezen, maar om te bepalen welke retrievalarchitectuur het beste aansluit bij de praktische eisen van de toepassing. Een retriever rechtsboven in de grafiek combineert hoge retrievalkwaliteit met hoge throughput en vormt daardoor meestal de beste algemene balans voor real-time RAG-systemen. Een retriever kan echter bewust meer querytijd gebruiken om betere ranking of hogere retrievaldekking te behalen. In productieomgevingen moet daarom rekening worden gehouden met latency, schaalbaarheid, hardwarekosten en onderhoudscomplexiteit. Een iets lagere retrievalscore kan acceptabel zijn wanneer een systeem aanzienlijk sneller, goedkoper of eenvoudiger schaalbaar blijft.

Rank Distribution

Retriever	Queries	Rank 1	Rank 2-3	Rank 4-10	Rank 11+	Avg First Rank
BM25Retriever (bm25)	99	63	14	12	9	2.9798
BM25Retriever (bm25) + (rerank)	99	90	4	2	2	1.3939
SpladeRetriever (splade)	95	60	17	14	3	2.2737
SpladeRetriever (splade) + (rerank)	95	85	6	3	0	1.2421
DenseRetrieverSentenceBert (dense)	102	80	14	6	1	1.5196
DenseRetrieverSentenceBert (dense) + (rerank)	102	92	6	2	1	1.2647
HybridRetriever (hybrid)	102	79	15	6	1	1.6275
HybridRetriever (hybrid) + (rerank)	102	93	5	2	1	1.2451
LinearRagRetriever (linear)	102	81	12	7	1	1.5588
LinearRagRetriever (linear) + (rerank)	102	93	5	3	0	1.1569



Uitleg rank distribution: Rank distribution laat zien op welke positie het eerste relevante document verschijnt voor iedere query. In plaats van alleen een algemene retrievalscore te tonen, maakt deze verdeling zichtbaar hoe snel bruikbare informatie beschikbaar komt voor de RAG-pipeline. Rank 1 betekent dat direct het eerste resultaat relevant is. Rank 2-3 betekent dat relevante informatie nog steeds zeer vroeg wordt gevonden. Rank 4-10 geeft aan dat de retriever dieper in de ranking moet zoeken voordat bruikbaar bewijs verschijnt. Rank 11+ betekent dat relevante informatie pas laat wordt gevonden, waardoor de kans groter wordt dat het taalmodel minder sterke of minder gefocuste context ontvangt. De gemiddelde first rank vat samen hoe vroeg het eerste relevante document gemiddeld verschijnt; lagere waarden duiden op sterkere rankingprestaties.

Praktische interpretatie: Een retriever met veel Rank 1 en Rank 2-3 resultaten levert doorgaans directer bruikbaar bewijs voor generatie. Dit is vooral waardevol wanneer het contextvenster beperkt is of wanneer de generator sterk afhankelijk is van de eerste passages in de prompt. Een groter aandeel Rank 4-10 kan nog steeds bruikbaar zijn, maar betekent dat relevante informatie minder prominent wordt aangeboden. Een groot Rank 11+ aandeel wijst vaak op zwakkere rankingkwaliteit of op een grotere behoefte aan reranking, hybride retrieval of betere queryverwerking. Rank distribution helpt daarom niet alleen te bepalen of relevante documenten worden gevonden, maar vooral hoe snel en hoe zichtbaar deze documenten voor de generator beschikbaar zijn.

Scalability Estimates

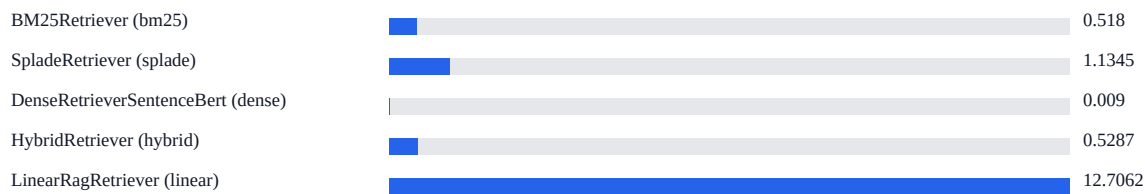
Retriever	Target Docs	Est. Index Time	Target Queries	Est. Query Time	Est. Queries/Sec	Est. Storage
BM25Retriever (bm25)	1,000,000	28.51min	100,000	8.55min	194.97	771.84MB
SpladeRetriever (splade)	1,000,000	6.05h	100,000	18.72min	89.0287	9.95GB
DenseRetrieverSentenceBert (dense)	1,000,000	1.5104s	100,000	8.9028s	11232.4526	4.00GB
HybridRetriever (hybrid)	1,000,000	28.53min	100,000	8.72min	191.0499	4.75GB
LinearRagRetriever (linear)	1,000,000	1.42h	100,000	3.49h	7.9489	881.85GB



Total Index Time (seconds) (2594 docs)



Total Query Time (seconds) (101 queries)

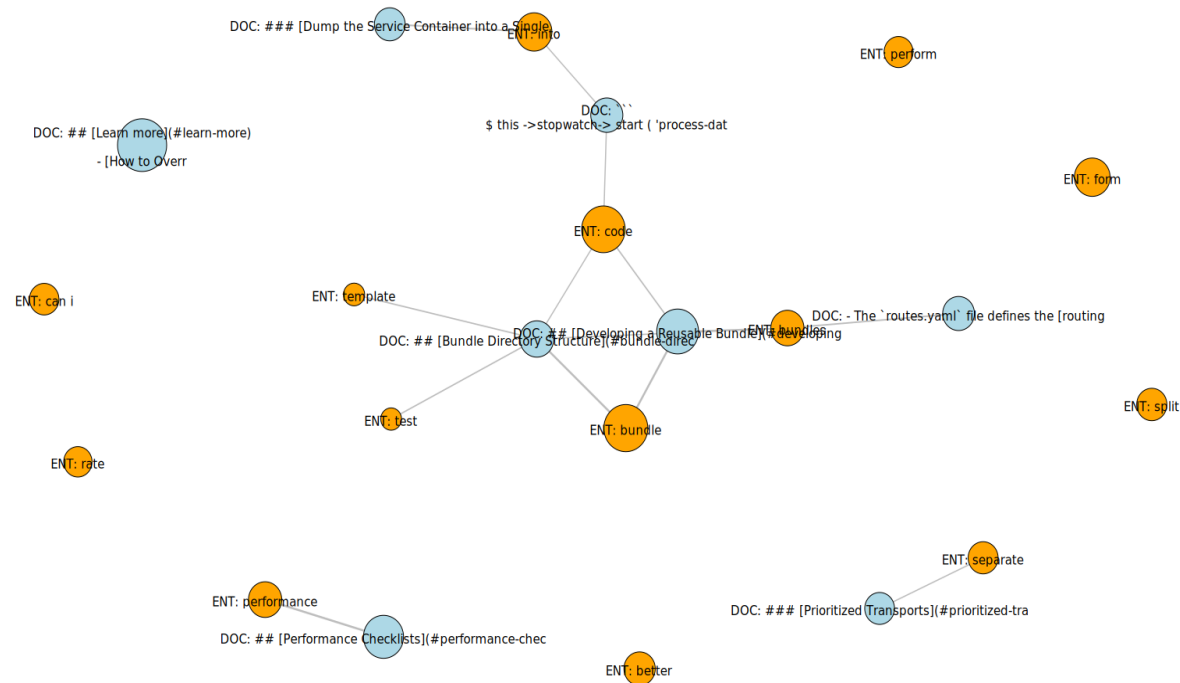


Rerank Time (seconds) (20 - docs / 101 - queries)



Query Graph - LinearRagRetriever (linear)

Query: how can I split code into separate bundles for better performance



Retrieved Documents

Rank	Document	Score	Text
1	9ab9fe6c-a734-4719-9731-bb95b2f94f2a	0.0164	## [Learn more](#learn-more) - [How to Override any Part of a Bundle](bundles/override.html) - [Best Practices for Reusable Bun...
2	b07febcb-5230-49ae-b5c6-7a2be047f2ae	0.0161	## [Developing a Reusable Bundle](#developing-a-reusable-bundle) Bundles are meant to be reusable pieces of code that live inde...
3	3053e751-a4af-4260-87ea-a83354a5649c	0.0159	## [Performance Checklists](#performance-checklists) Use these checklists to verify that your application and server are config...
4	d717579f-6a2f-4da6-8126-e0a8ca82efbc	0.0156	## [Bundle Directory Structure](#bundle-directory-structure) The directory structure of a bundle is meant to help to keep code...
5	831562a6-5c07-4639-81e9-c0eafae5dabb	0.0154	``` \$ this ->stopwatch-> start ('process-data-records' , 'export'); foreach (\$ records as \$ record) { // ... some code goes...

6	9ea60115-3512-4822-9414-40e08b1cd242	0.0152	- The `routes.yaml` file defines the [routing configuration](routing.html) ; - The `services.yaml` file configures the services...
7	95ae58dd-85ec-42f3-bfdb-2824187fb0b3	0.0149	### [Dump the Service Container into a Single File](#dump-the-service-container-into-a-single-file) Symfony compiles the [servi...
8	502fde83-52ef-4a52-8f44-4a9fccee50c1	0.0147	### [Prioritized Transports](#prioritized-transports) Use separate transports for message types with different latency requirem...

BM25Retriever (bm25)

Configuration

Retriever Type	bm25
Topk	20
Language	english
Index Name	raccoon-bm25-test
Content Field	content
Metadata Field	metadata
Refresh On Write	yes
Timeout	120

Result Summary

Queries	101
Total Results	2,020
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	7.7833
Avg Score	13.8247
Max Score	41.626

Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.571	0.4869	0.4317	0.5017	0.5606
MAP	0.1366	0.2743	0.3194	0.3763	0.4084
Recall	0.1366	0.3	0.3792	0.5129	0.6584
Precision	0.6139	0.4719	0.3624	0.2465	0.1579

Retriever Metrics

Index Time / Indexing / Time In Seconds	4.437
Index Time / Indexing / Documents	2,594
Index Time / Indexing / Docs Per Second	584.626
Index Time / Indexing / Backend	elasticsearch
Index Time / Indexing / Storage Size Bytes	2,099,396
Index Time / Indexing / Storage Size Mb	2.0021
Query Time / Search / Time In Seconds	0.518
Query Time / Search / Time Per Query In Seconds	0.0051
Query Time / Search / Queries Per Second	194.97
Query Time / Search / Queries	101

Query Time / Search / Top K	20
Query Time / Search / Backend	elasticsearch
Query Time / Search / Total Results	2,020
Query Time / Search / Avg Results Per Query	20
Query Time / Search / Min Results Per Query	20
Query Time / Search / Max Results Per Query	20
Query Time / Search / Min Score	7.7833
Query Time / Search / Max Score	41.626
Query Time / Search / Avg Score	13.8247

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	101
Total Pairs	2,020
Total Batches	202
Total Input Tokens	501,366
Total Wall Time Sec	8.4779
Avg Time Query Sec	0.0839
Avg Time Batch Sec	0.0419
Pairs Per Sec	238.27
Queries Per Sec	11.91
Tokens Per Sec	59137.86
Peak Gpu Memory Gb	1.21

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.8366	0.7049	0.6117	0.6622	0.6754
MAP	0.1916	0.4134	0.4969	0.5436	0.5521
Recall	0.1916	0.4243	0.5282	0.6287	0.6584
Precision	0.8713	0.6733	0.505	0.301	0.1579

Sample Top Results

Query q0: how can I split code into separate bundles for better performance

1. Document b57212bb-dfee-4650-9f72-57fa2cb8c67d | score 14.8736

[Can I Lint and Format My Code?](#can-i-lint-and-format-my-code) Not with AssetMapper, but you can install [kocal/biome-js-bundle](https://github.com/Kocal/BiomeJsBundle) in your project to lint and format your front-end assets. It's much faster than alternatives like Prettier and requires no configuration to handle your JavaScript, TypeScript and CSS files.

2. Document eaeae399-750e-4d2d-afdb-3116f6a4135b | score 13.0967

[Can I Use it with JSX or Vue?](#can-i-use-it-with-jsx-or-vue) Probably not. And if you're writing an application in React, Svelte or another frontend framework, you'll probably be better off using *their* tools directly. JSX *can* be compiled directly to a native JavaScript file but if you're using a lot of JSX, you'll probably want to use a tool like [Encore](../frontend.html#frontend-webpack-encore) . See the [UX React Documentation](https://symfony.com/bundles/ux-react/current/index.h...

3. Document b6bf40b2-b1d0-4518-8e20-b2abda8d5422 | score 11.889

[How to Set Up Before and After Filters](#how-to-set-up-before-and-after-filters) It is quite common in web application development to need some logic to be performed right before or directly after your controller actions acting as filters or hooks. Some web frameworks define methods like `preExecute()` and `postExecute()` , but there is no such thing in Symfony. The good news is that there is a much better way to interfere with the Request & Response process using the [EventDispatcher...

Sample Top Rerank Results

Query q0: how can I split code into separate bundles for better performance

1. Document e9563c4d-fc97-4a21-8397-149ded4f7458 | score -4.9336

[Getting Started](#getting-started) - [Installation](installation.html) - [Using Webpack Encore](simple-example.html) ## [Adding more Features](#adding-more-features) - [CSS Preprocessors: Sass, LESS, etc.](css-preprocessors.html) - [PostCSS and autoprefixing](postcss.html) - [Enabling React.js](reactjs.html) - [Enabling Vue.js (vue-loader)](vuejs.html) - [Copying & Referencing Images with Webpack Encore](copy-files.html) - [Configuring Babel](babel.html) - [Source maps](sourcemaps.htm...

2. Document b57212bb-dfee-4650-9f72-57fa2cb8c67d | score -4.9492

[Can I Lint and Format My Code?](#can-i-lint-and-format-my-code) Not with AssetMapper, but you can install [kocal/biome-js-bundle](https://github.com/Kocal/BiomeJsBundle) in your project to lint and format your front-end assets. It's much faster than alternatives like Prettier and requires no configuration to handle your JavaScript, TypeScript and CSS files.

3. Document 3393fb78-d2e8-4ae2-be75-8c5d471c794d | score -5.0273

[Including Templates](#including-templates) If certain Twig code is repeated in several templates, you can extract it into a single "template fragment" and include it in other templates. Imagine that the following code to display the user information is repeated in several places: `` {# templates/blog/index.html.twig #} {# ... #} <div class = "user-profile" > <p > {{ user.fullName }} - {{ user.email }} </p > </...`

SpladeRetriever (splade)

Configuration

Retriever Type	splade
Topk	20
Language	dutch
Index Name	raccoon-splade-test
Content Field	content
Metadata Field	metadata
Refresh On Write	yes
Timeout	120
Max Length	512
Device	cuda
Batch Size	8

Result Summary

Queries	101
Total Results	2,020
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	6.4248
Avg Score	10.8045

Max Score	20.3686
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Retriever Metrics

Index Time / Indexing / Time In Seconds	56.4633
Index Time / Indexing / Documents	2,594
Index Time / Indexing / Docs Per Second	45.9414
Index Time / Indexing / Backend	elasticsearch
Index Time / Indexing / Model	sparse-encoder/splade-robbert-dutch-base-v1
Index Time / Indexing / Device	cuda
Index Time / Indexing / Storage Size Bytes	27,722,967
Index Time / Indexing / Storage Size Mb	26.4387
Index Time / Indexing / Max Features	1,024
Query Time / Search / Time In Seconds	1.1345
Query Time / Search / Time Per Query In Seconds	0.0112
Query Time / Search / Queries Per Second	89.0287
Query Time / Search / Queries	101
Query Time / Search / Top K	20
Query Time / Search / Backend	elasticsearch
Query Time / Search / Model	sparse-encoder/splade-robbert-dutch-base-v1
Query Time / Search / Device	cuda
Query Time / Search / Total Results	2,020
Query Time / Search / Avg Results Per Query	20
Query Time / Search / Min Results Per Query	20
Query Time / Search / Max Results Per Query	20
Query Time / Search / Min Score	6.4248
Query Time / Search / Max Score	20.3686
Query Time / Search / Avg Score	10.8045
Query Time / Search / Max Features	1,024

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	101
Total Pairs	2,020
Total Batches	202
Total Input Tokens	554,515
Total Wall Time Sec	8.3586
Avg Time Query Sec	0.0827
Avg Time Batch Sec	0.0413
Pairs Per Sec	241.67
Queries Per Sec	12.08
Tokens Per Sec	66341.04
Peak Gpu Memory Gb	1.82

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.7855	0.6579	0.566	0.6022	0.6115
MAP	0.1762	0.3802	0.4495	0.4846	0.49
Recall	0.1762	0.3931	0.4842	0.5584	0.5812
Precision	0.8119	0.6271	0.4673	0.2693	0.1396

Sample Top Results

Query q0: how can I split code into separate bundles for better performance

1. Document 502fde83-52ef-4a52-8f44-4a9fccee50c1 | score 8.1307

[Prioritized Transports](#prioritized-transports) Use separate transports for message types with different latency requirements, failure modes, or retry windows. When multiple message types share the same transport, a slow or failing handler for one type can delay all others in the same queue. For example, if catalog sync messages and payment confirmations are routed to the same transport, a slow product feed handler could delay payment confirmations for customers going through checkout....

2. Document 2c4b8bb0-71ca-4e09-a870-c6dc47246bfc | score 8.1205

[Path Manipulation Utilities](#path-manipulation-utilities) Dealing with file paths usually involves some difficulties: - Platform differences: file paths look different on different platforms. UNIX file paths start with a slash ("/"), while Windows file paths start with a system drive ("C:"). UNIX uses forward slashes, while Windows uses backslashes by default. However, Windows also accepts forward slashes, so both types of separators generally work. - Absolute/relative paths: web applica...

3. Document 97d2772c-6852-4a95-bfc9-163f9aed0116 | score 7.7015

[Design Principles](#design-principles) - Contracts are split by domain, each into their own sub-namespaces; - Contracts are small and consistent sets of PHP interfaces, traits, normative docblocks and reference test suites when applicable, ...; - Contracts must have a proven implementation to enter this repository; - Contracts must be backward compatible with existing Symfony components. Packages that implement specific contracts should list them in the `provide` section of their `compose...

Sample Top Rerank Results

Query q0: how can I split code into separate bundles for better performance

1. Document ac58f573-e300-4d0a-9b76-0a71eef46fdd | score -4.7539

[Use the OPcache class preloading](#use-the-opcache-class-preloading) OPcache can compile and load classes at start-up and make them available to all requests until the server is restarted, improving performance significantly. During container compilation (e.g. when running the `cache:clear` command), Symfony generates a file with the list of classes to preload in the `var/cache/` directory. Rather than use this file directly, use the `config/preload.php` file that is created when [using...

2. Document 95ae58dd-85ec-42f3-bfdb-2824187fb0b3 | score -4.7891

[Dump the Service Container into a Single File](#dump-the-service-container-into-a-single-file) Symfony compiles the [service container](service_container.html) into multiple small files by default. Set this parameter to `true` to compile the entire container into a single file, which could improve performance when using PHP [class preloading](https://php.net/opcache.preloading.php) : `YAML PHP `` # config/services.yaml parameters: #container.dumper.inline_factories: true ``` Tip Th...

3. Document 7eeb5a53-f0d8-4344-8962-9a9480ef9bcd | score -5.3711

is where you need to write your logic. It will be called for each chunk yielded by the underlying client. A ``$passthru`` that does nothing would just ``yield` `` $chunk; ``` . You could also yield a modified chunk, split the chunk into many ones by yielding several times, or even skip a chunk altogether by issuing a ``return`` instead of yielding. In order to control the stream, the chunk passthru receives an `[AsyncContext]`(<https://github.com/symfony/symfony/blob/8.1/src/Symfony/Component/HttpC...>

DenseRetrieverSentenceBert (dense)

Configuration

Retriever Type	dense
Topk	20
Model Id	snowflake/snowflake-arctic-embed-l-v2.0
Max Length	512
Device	cuda
Query Prompt Name	query
Passage Prompt Name	document
Normalize Embeddings	no
Batch Size	128
Corpus Chunk Size	50,000
Query Chunk Size	1,024
Show Progress Bar	yes

Result Summary

Queries	101
Total Results	2,020
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	0.2484
Avg Score	0.5006
Max Score	0.8164

Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.7558	0.6725	0.6205	0.7091	0.7701
MAP	0.1772	0.3874	0.4959	0.5826	0.628
Recall	0.1772	0.4153	0.5688	0.7446	0.8951
Precision	0.802	0.6535	0.5445	0.3584	0.2168

Retriever Metrics

Index Time / Encoding / Time In Seconds	0.0039
Index Time / Encoding / Documents	2,594
Index Time / Encoding / Queries	101
Index Time / Encoding / Shards	1

Index Time / Encoding / Embedding Dim	1,024
Index Time / Encoding / Size In Bytes	11,140,835
Index Time / Encoding / Size In Mb	10.6247
Index Time / Encoding / Docs Per Second	662083.1235
Query Time / Search / Time In Seconds	0.009
Query Time / Search / Time Per Query In Seconds	0.0001
Query Time / Search / Queries Per Second	11232.4526
Query Time / Search / Queries	101
Query Time / Search / Top K	20
Query Time / Search / Score Function	dot
Query Time / Search / Use Faiss	no
Query Time / Search / Backend	torch
Query Time / Search / Total Results	2,020
Query Time / Search / Avg Results Per Query	20
Query Time / Search / Min Results Per Query	20
Query Time / Search / Max Results Per Query	20
Query Time / Search / Min Score	0.2484
Query Time / Search / Max Score	0.8164
Query Time / Search / Avg Score	0.5006

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	101
Total Pairs	2,020
Total Batches	202
Total Input Tokens	497,956
Total Wall Time Sec	8.2656
Avg Time Query Sec	0.0818
Avg Time Batch Sec	0.0409
Pairs Per Sec	244.39
Queries Per Sec	12.22

Tokens Per Sec	60244.29
Peak Gpu Memory Gb	3.94

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.901	0.7669	0.7013	0.77	0.8213
MAP	0.2015	0.4543	0.5979	0.6675	0.7031
Recall	0.2015	0.4639	0.6347	0.7703	0.8951
Precision	0.9208	0.7393	0.6139	0.3743	0.2168

Sample Top Results

Query q0: how can I split code into separate bundles for better performance

1. Document b07febcb-5230-49ae-b5c6-7a2be047f2ae | score 0.4644

[Developing a Reusable Bundle](#developing-a-reusable-bundle) Bundles are meant to be reusable pieces of code that live independently from any particular Symfony application. However, a bundle cannot run on its own: it must be registered inside an application to execute its code. This can be a bit challenging during development. When working on a bundle in its own repository, there's no Symfony application around it, so you need a way to test your changes inside a real application environm...

2. Document 9ab9fe6c-a734-4719-9731-bb95b2f94f2a | score 0.4292

[Learn more](#learn-more) - [How to Override any Part of a Bundle](bundles/override.html) - [Best Practices for Reusable Bundles](bundles/best_practices.html) - [How to Create Friendly Configuration for a Bundle](bundles/configuration.html) - [How to Load Service Configuration inside a Bundle](bundles/extension.html) - [How to Simplify Configuration of Multiple Bundles](bundles/prepend_extension.html)

3. Document 18a7775c-6bb4-420e-8c1f-3284c436883b | score 0.3905

[Linking an Already Published Bundle](#linking-an-already-published-bundle) If your bundle is already public (for example, it's published on Packagist), you can still develop it locally while testing it inside a Symfony application. In your application, replace the installed bundle with a symlink to your local development copy. For example, if your bundle is installed under `vendor/acme/blog-bundle/` and your local copy is in `~/Projects/AcmeBlogBundle/` : `` \$ rm -rf vendor/acme/blog-bu...

Sample Top Rerank Results

Query q0: how can I split code into separate bundles for better performance

1. Document 9ccdbec1-f107-435b-9a7a-157e8dae1d75 | score -3.6504

[Using a Local Path Repository](#using-a-local-path-repository) If your bundle hasn't been published yet (for example, it's not available on Packagist), you can point Composer to your local bundle directory from any Symfony application you use for testing. Edit the `composer.json` file of your application and add this: `` { "repositories" : [ { "type" : "path" , "url" : "/path/to/your/AcmeBlogBundle" } ], "require" : { "acme/blog-bundle" : "\*" } } `` Then, in your application, install t...

2. Document d717579f-6a2f-4da6-8126-e0a8ca82efbc | score -4.7305

[Bundle Directory Structure](#bundle-directory-structure) The directory structure of a bundle is meant to help to keep code consistent between all Symfony bundles. It follows a set of conventions, but is flexible to be adjusted if needed: `assets/` Contains the web asset sources like JavaScript and TypeScript files, CSS and Sass files, but also images and other assets related to the bundle that are not in `public/` (e.g. Stimulus controllers). `config/` Houses configuration, including rout...

3. Document 95ae58dd-85ec-42f3-bfdb-2824187fb0b3 | score -4.7891

[Dump the Service Container into a Single File](#dump-the-service-container-into-a-single-file) Symfony compiles the [service container](service_container.html) into multiple small files by default. Set this parameter to `true` to compile the entire container into a single file, which could improve performance when using PHP [class preloading](https://php.net/opcache.preloading.php) : YAML PHP `` # config/services.yaml parameters: # ... .container.dumper.inline\_factories: true `` Tip Th...

HybridRetriever (hybrid)

Configuration

Retriever Type	hybrid
K	60
Retrievers	bm25:0.3, dense:0.7

Result Summary

Queries	101
Total Results	2,020
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	0.0087
Avg Score	0.0116
Max Score	0.0164

Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.7426	0.655	0.5884	0.7087	0.7613
MAP	0.1708	0.3789	0.4622	0.5775	0.6165
Recall	0.1708	0.4099	0.5317	0.7644	0.8951
Precision	0.7822	0.6436	0.505	0.3683	0.2168

Retriever Metrics

Query Time / Search / Time In Seconds	0.5287
Query Time / Search / Retriever Time In Seconds	0.527
Query Time / Search / Rrf Time In Seconds	0.0016
Query Time / Search / Wrapper Time In Seconds	0.0018
Query Time / Search / Execution Mode	sequential
Query Time / Search / Time Per Query In Seconds	0.0052
Query Time / Search / Queries Per Second	191.0499
Query Time / Search / Queries	101
Query Time / Search / Top K	20
Query Time / Search / Retrievers	2
Query Time / Search / Rrf K	60
Query Time / Search / Total Results	2,020
Query Time / Search / Avg Results Per Query	20
Query Time / Search / Min Results Per Query	20
Query Time / Search / Max Results Per Query	20
Query Time / Search / Min Score	0.0087
Query Time / Search / Max Score	0.0164
Query Time / Search / Avg Score	0.0116
Index Time / Indexing / Time In Seconds	4.4409

Index Time / Indexing / Documents	5,188
Index Time / Indexing / Docs Per Second	1168.2205
Index Time / Indexing / Retrievers	2
Index Time / Indexing / Size In Mb	12.6269

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	101
Total Pairs	2,020
Total Batches	202
Total Input Tokens	497,956
Total Wall Time Sec	8.2762
Avg Time Query Sec	0.0819
Avg Time Batch Sec	0.0409
Pairs Per Sec	244.07
Queries Per Sec	12.2
Tokens Per Sec	60167.32
Peak Gpu Memory Gb	3.94

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.901	0.7669	0.7013	0.77	0.8213
MAP	0.2015	0.4543	0.5979	0.6675	0.7031
Recall	0.2015	0.4639	0.6347	0.7703	0.8951
Precision	0.9208	0.7393	0.6139	0.3743	0.2168

Sample Top Results

Query q0: how can I split code into separate bundles for better performance

1. Document b07febcb-5230-49ae-b5c6-7a2be047f2ae | score 0.0155

[Developing a Reusable Bundle](#developing-a-reusable-bundle) Bundles are meant to be reusable pieces of code that live independently from any particular Symfony application. However, a bundle cannot run on its own: it must be registered inside an application to execute its code. This can be a bit challenging during development. When working on a bundle in its own repository, there's no Symfony application around it, so you need a way to test your changes inside a real application environm...

2. Document 9ab9fe6c-a734-4719-9731-bb95b2f94f2a | score 0.0155

[Learn more](#learn-more) - [How to Override any Part of a Bundle](bundles/override.html) - [Best Practices for Reusable Bundles](bundles/best_practices.html) - [How to Create Friendly Configuration for a Bundle](bundles/configuration.html) - [How to Load Service Configuration inside a Bundle](bundles/extension.html) - [How to Simplify Configuration of Multiple Bundles](bundles/prepend_extension.html)

3. Document e9563c4d-fc97-4a21-8397-149ded4f7458 | score 0.0155

[Getting Started](#getting-started) - [Installation](installation.html) - [Using Webpack Encore](simple-example.html) ## [Adding more Features](#adding-more-features) - [CSS Preprocessors: Sass, LESS, etc.](css-preprocessors.html) - [PostCSS and autoprefixing](postcss.html) - [Enabling React.js](reactjs.html) - [Enabling Vue.js (vue-loader)](vuejs.html) - [Copying & Referencing Images with Webpack

Sample Top Rerank Results

Query q0: how can I split code into separate bundles for better performance

1. Document 9ccdbec1-f107-435b-9a7a-157e8dae1d75 | score -3.6504

[Using a Local Path Repository](#using-a-local-path-repository) If your bundle hasn't been published yet (for example, it's not available on Packagist), you can point Composer to your local bundle directory from any Symfony application you use for testing. Edit the `composer.json` file of your application and add this: `` { "repositories" : [ { "type" : "path" , "url" : "/path/to/your/AcmeBlogBundle" } ], "require" : { "acme/blog-bundle" : "\*" } } `` Then, in your application, install t...

2. Document d717579f-6a2f-4da6-8126-e0a8ca82efbc | score -4.7305

[Bundle Directory Structure](#bundle-directory-structure) The directory structure of a bundle is meant to help to keep code consistent between all Symfony bundles. It follows a set of conventions, but is flexible to be adjusted if needed: `assets/` Contains the web asset sources like JavaScript and TypeScript files, CSS and Sass files, but also images and other assets related to the bundle that are not in `public/` (e.g. Stimulus controllers). `config/` Houses configuration, including rout...

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[Dump the Service Container into a Single File](#dump-the-service-container-into-a-single-file) Symfony compiles the [service container](service_container.html) into multiple small files by default. Set this parameter to `true` to compile the entire container into a single file, which could improve performance when using PHP [class preloading](https://php.net/opcache.preloading.php) : YAML PHP `` # config/services.yaml parameters: # ... .container.dumper.inline\_factories: true `` Tip Th...

LinearRagRetriever (linear)

Configuration

Retriever Type	linear
Embedding Model Name	snowflake/snowflake-arctic-embed-l-v2.0
Spacy Model Name	nl_core_news_sm
Dataset Name	symfony_beir_semantic
Device	cuda
Max Seq Length	512
Embed Batch Size	128
Ner Batch Size	64
Retrieval Top K	20
Dense Candidate K	500
Bm25 Candidate K	500
Graph Candidate K	500
Local Graph Dense Seed K	200
Local Graph Bm25 Seed K	200
Local Graph Sentence Seed K	50
Local Graph Concept Seed K	50
Dense Rrf Weight	0
Bm25 Rrf Weight	0
Graph Rrf Weight	1
Rrf K	60
Min Concept Len	4
Max Concept Words	4
Min Concept Df	2
Max Concept Df Ratio	0.1
Ppr Damping	0.6

Ppr Max Result Docs	500
Export Query Graph	yes
Use Gpu For Spacy	yes

Result Summary

Queries	101
Total Results	2,020
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	0.0125
Avg Score	0.0143
Max Score	0.0164

Retrieval Metrics

Metric	@1	@3	@5	@10	@50	@100
NDCG	0.7475	0.6604	0.5896	0.6952	0.7529	0.7529
MAP	0.1772	0.3822	0.4617	0.563	0.6051	0.6051
Recall	0.1772	0.4079	0.5267	0.7292	0.8733	0.8733
Precision	0.802	0.637	0.501	0.3515	0.0846	0.0423

Retriever Metrics

Index Time / Indexing / Time In Seconds	13.2822
Index Time / Indexing / Documents	2,594
Index Time / Indexing / Concepts	5,255
Index Time / Indexing / Sentences	26,480
Index Time / Indexing / Concepts Per Document	2.0258
Index Time / Indexing / Sentences Per Document	10.2082
Index Time / Indexing / Passage Concept Links	55,496
Index Time / Indexing / Concept Sentence Links	54,784
Index Time / Indexing / Sentence Passage Links	41,217
Index Time / Indexing / Embedding Dim	1,024
Index Time / Indexing / Size In Bytes	2,456,211,643
Index Time / Indexing / Size In Mb	2342.426
Index Time / Indexing / Docs Per Second	195.2992
Index Time / Indexing / Cache Loaded	yes
Index Time / Indexing / Cache Path	data/processed/symfony/symfony_beir_semantic/linear_rag_cache/snowf...

Index Time / Indexing / Embedding Model	snowflake/snowflake-arctic-embed-l-v2.0
Index Time / Indexing / Spacy Model	nl_core_news_sm
Index Time / Indexing / Backend	faiss
Index Time / Indexing / Phase Times / Model Load Time In Seconds	7.8804
Index Time / Indexing / Phase Times / Cache Load Time In Seconds	5.4017
Query Time / Search / Time In Seconds	12.7062
Query Time / Search / Time Per Query In Seconds	0.1258
Query Time / Search / Queries Per Second	7.9489
Query Time / Search / Queries	101
Query Time / Search / Top K	20
Query Time / Search / Dense Candidate K	500
Query Time / Search / Bm25 Candidate K	500
Query Time / Search / Graph Candidate K	500
Query Time / Search / Dense Rrf Weight	0
Query Time / Search / Bm25 Rrf Weight	0

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	101
Total Pairs	2,020
Total Batches	202
Total Input Tokens	510,331
Total Wall Time Sec	8.2209
Avg Time Query Sec	0.0814
Avg Time Batch Sec	0.0407
Pairs Per Sec	245.72
Queries Per Sec	12.29
Tokens Per Sec	62077.46
Peak Gpu Memory Gb	7.11

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.901	0.7692	0.6968	0.7581	0.8091
MAP	0.2015	0.4543	0.5868	0.6476	0.6835
Recall	0.2015	0.4639	0.6267	0.7495	0.8733
Precision	0.9208	0.7393	0.6059	0.3644	0.2114

Sample Top Results

Query q0: how can I split code into separate bundles for better performance

1. Document 9ab9fe6c-a734-4719-9731-bb95b2f94f2a | score 0.0164

[Learn more](#learn-more) - [How to Override any Part of a Bundle](bundles/override.html) - [Best Practices for Reusable Bundles](bundles/best_practices.html) - [How to Create Friendly Configuration for a Bundle](bundles/configuration.html) - [How to Load Service Configuration inside a Bundle](bundles/extension.html) - [How to Simplify Configuration of Multiple Bundles](bundles/prepend_extension.html)

2. Document b07febcb-5230-49ae-b5c6-7a2be047f2ae | score 0.0161

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3. Document 3053e751-a4af-4260-87ea-a83354a5649c | score 0.0159

[Performance Checklists](#performance-checklists) Use these checklists to verify that your application and server are configured for maximum performance: - **Symfony Application Checklist** : 1. [Restrict the number of locales enabled in the application](performance.html#performance-enabled-locales) - **Production Server Checklist** : 1. [Dump the service container into a single file](performance.html#performance-service-container-single-file) 2. [Use the OPcache bytecode cache](performanc...

Sample Top Rerank Results

Query q0: how can I split code into separate bundles for better performance

1. Document 85690100-0283-460c-96fc-c7a3219b3153 | score -3.3945

[Installation](#installation) Before creating your first test, install `symfony/test-pack`, which installs some other packages needed for testing (such as `phpunit/phpunit`): `` \$ composer require --dev symfony/ test -pack `` After the library is installed, try running PHPUnit: `` \$ php bin/phpunit `` This command automatically runs your application tests. Each test is a PHP class ending with "Test" (e.g. `BlogControllerTest`) that lives in the `tests/` directory of your application...

2. Document 9ccdbec1-f107-435b-9a7a-157e8dae1d75 | score -3.6504

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